3. Equations Reducible to Homogeneous Form

Sometimes, a differential equation looks almost homogeneous but contains constant terms that break the symmetry. These equations usually describe the intersection of non-homogeneous lines. We can "reduce" them to a homogeneous form (or directly to variable separable form) using specific substitutions.

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1. Concept Overview

This method applies to differential equations of the form:

$$\frac{dy}{dx} = \frac{ax + by + c}{Ax + By + C}$$

Here, the constants c and C prevent the equation from being homogeneous. To solve this, we compare the ratios of the coefficients of x and y.

There are two distinct cases:

Case 1: Intersecting Lines ($\frac{a}{A} \neq \frac{b}{B}$)

If the ratios are **not equal**, the lines represented by the numerator and denominator intersect at a specific point (h, k). * **Substitution:** We shift the origin to that intersection point. * $x = X + h$ * $y = Y + k$ * $dy = dY, dx = dX$ * We find h and k by solving the system of equations $ax+by+c = 0$ and $Ax+By+C = 0$. * This removes the constants c and C, resulting in a standard Homogeneous DE in variables X and Y.

Case 2: Parallel Lines ($\frac{a}{A} = \frac{b}{B}$)

If the ratios **are equal**, the lines are parallel. This means the terms $(ax + by)$ and $(Ax + By)$ are scalar multiples of each other. * **Substitution:** We substitute the common linear expression with a new variable z. * Let $z = ax + by$ * Differentiate with respect to x to find $\frac{dz}{dx}$. * This immediately reduces the DE to a **Variable Separable** form.

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2. Solved Questions

Here is a step-by-step solution for a **Case 2** problem found in the lecture notes.

Question 1 (Case 2: Equal Ratios)

Solve: $(x + 2y)(dx - dy) = dx + dy$

Solution:

Step 1: Rewrite in Standard Form First, rearrange the equation to isolate $\frac{dy}{dx}$. Expand the terms:

$$x\,dx - x\,dy + 2y\,dx - 2y\,dy = dx + dy$$

Group dx and dy terms:

$$(x + 2y)\,dx - dx = (x + 2y)\,dy + dy$$

$$(x + 2y - 1)\,dx = (x + 2y + 1)\,dy$$

$$\frac{dy}{dx} = \frac{x + 2y - 1}{x + 2y + 1}$$

Step 2: Check Ratios Compare coefficients of x and y in the numerator ($a = 1, b = 2$) and denominator ($A = 1, B = 2$).

$$\frac{a}{A} = \frac{1}{1} = 1, \quad \frac{b}{B} = \frac{2}{2} = 1$$

Since $\frac{a}{A} = \frac{b}{B}$, this is **Case 2**.

Step 3: Apply Substitution Let the common term be z.

$$z = x + 2y$$

Differentiate with respect to x:

$$\frac{dz}{dx} = 1 + 2\frac{dy}{dx} \implies \frac{dy}{dx} = \frac{1}{2}\left(\frac{dz}{dx} - 1\right)$$

Step 4: Substitute into the DE Replace $\frac{dy}{dx}$ and $(x + 2y)$ in the standard form equation:

$$\frac{1}{2}\left(\frac{dz}{dx} - 1\right) = \frac{z-1}{z+1}$$

Multiply by 2:

$$\frac{dz}{dx} - 1 = \frac{2(z-1)}{z+1} = \frac{2z-2}{z+1}$$

Add 1 to both sides:

$$\frac{dz}{dx} = \frac{2z-2}{z+1} + 1$$

Find common denominator:

$$\frac{dz}{dx} = \frac{2z-2+(z+1)}{z+1} = \frac{3z-1}{z+1}$$

Step 5: Separate Variables and Integrate

$$\frac{z+1}{3z-1}dz = dx$$

Integrate both sides:

$$\int \frac{z+1}{3z-1}dz = \int dx$$

Integration Tip: To integrate the left side, make the numerator resemble the derivative of the denominator or perform long division. Multiply and divide by 3 inside the integral for easier manipulation:

$$\frac{1}{3}\int \frac{3z+3}{3z-1}dz = x + C$$

$$\frac{1}{3}\int \frac{(3z-1)+4}{3z-1}dz = x + C$$

$$\frac{1}{3}\int \left(1 + \frac{4}{3z-1}\right)dz = x + C$$

$$\frac{1}{3}\left[z + 4 \cdot \frac{\ln|3z-1|}{3}\right] = x + C$$

Step 6: Final Substitution Multiply by 3 to clear the fraction:

$$z + \frac{4}{3}\ln|3z-1| = 3x + 3C$$

Substitute $z = x + 2y$ back:

$$(x + 2y) + \frac{4}{3} \ln |3(x + 2y) - 1| = 3x + C'$$

Rearrange for the final implicit solution:

$$2y - 2x + \frac{4}{3} \ln |3x + 6y - 1| = C'$$

Next Method: While Homogeneous and Reducible methods deal with ratios, many physical systems are modeled by **Linear Differential Equations**. Proceed to Method 4: Linear ODEs.